Cyclic Codes:

Definition:

A binary code is said to be a cyclic code if it cho exhibits the following properties

(a) Linearity (b) Cyclic

Linearity:

A code is said to be linear if sum of any two code words is also a codeword.

→ This properties state that cyclic codes are linear blockcodes.

Cyclic:

A code is said to be cyclic if any cyclic ~~code~~ shift of a code word results in the formation of another word. This Ex. shows the property of cyclic codes.

Ex: $x_0 \; x_1 \; x_2 \; \dots \; x_{n-1}$ ← Original codeword.

$$\Downarrow$$

New codeword → $x_{n-1} \; x_0 \; x_1 \; \dots \; x_{n-2}$

$$\Downarrow$$

New codeword → $x_{n-2} \; x_{n-1} \; x_0 \; \dots \; x_{n-3}$

$$\Downarrow$$

New codeword → $x_1 \; x_2 \; x_3 \; \dots \; x_0$

Shifted words. (New codewords).

Algebraic Structure of cyclic codes:

→ Code words are represented by a polynomial

$$X = (x_{n-1}, x_{n-2} \dots x_1, x_0)$$

→ Codeword can be represented by a polynomial of degree less than (or) equal to $(n-1)$ i.e,

$$X(p) = x_{n-1}p^{n-1} + x_{n-2}p^{n-2} + \dots x_1 p^1 + x_0$$

Where $X(p)$ = polynomial of degree $(n-1)$

p = arbitrary variable of the polynomial.

p - represents the positions of the codeword bits. $p^{n-1} \rightarrow$ MSB , $p^0 \rightarrow$ LSB

Reason's for polynomial Representation.

i) Algebraic Operations such as addition, multiplication, division, subtraction etc. becomes very simple

(ii) positions of bits are represented with the help of powers of p in a polynomial.

Generation of Non-systematic Code words:

Generator polynomial of cyclic code is represented by $G(p)$. It is used for the generation of cyclic code words.

→ The Generation of Codeword polynomial can be represented as:

$$X(p) = M(p) \cdot G(p).$$

Where $M(p) \rightarrow$ msg s/g polynomial of degree $\leq K$.

$G(p)$ = Generating polynomial of degree $(n-k)$

$$G(p) = 1 + g_1 p + g_2 p^2 + \cdots g_{n-k-1} p^{n-k-1} + p^{n-k}.$$

Generator polynomial is expressed as:

$$G(p) = 1 + \sum_{i=1}^{n-k-1} g_i p^i + p^{n-k}.$$

$$X_1(p) = M_1(p) \cdot G(p)$$

$$X_2(p) = M_2(p) \cdot G(p)$$

$$X_3(p) = M_3(p) \cdot G(p) \quad \ldots\ldots \text{ So on} \ldots$$

This is used for realizing the encoder for the cyclic Codes.

⇒ Degree of generator polynomial = no of parity bits (check bits)

Generation of systematic Code words.

Encoding process for a systematic (n,k) Cyclic Code involves 3 steps

(i) multiply msg polynomial $M(p)$ by p^{n-k} to get $p^{n-k}(M(p))$

(ii) Divide the sys shifted msg polynomial $p^{n-k}(M(p))$ by the generator polynomial $G(p)$ to obtain remainder $C(p)$

$$\frac{p^{n-k} M(p)}{G(p)} = Q(p) \oplus \frac{C(p)}{G(P)}$$

$Q(p)$ = Quotient.

(3) We add the remainder polynomial $C(p)$ to the shifted msg polynomial $p^{n-k} M(p)$ to obtain the code word polynomial $X(p)$.

$$\therefore X(p) = [\, p^{n-k} M(p)\,] \oplus C(p)$$

$X(p)$ is divisible by generator polynomial $G(p)$.

Q The Generator polynomial for (7,4) Cyclic Haming Code. is given by. $G(p) = 1 + p + p^3$.

Determine systemic & non systematic Code Vectors.

Sol^n: Non-Systematic Code Vectors:

(i) (7,4) cyclic Hamming code.

msg ~~Vector~~ Vector is 4.

$2^4 = 16$ msg words.

Consider.: ~~Example~~: $M = m_0 \; m_1 \; m_2 \; m_3$

$ 1 \;\; 0 \;\; 1 \;\; 0$

The msg. polynomial is given by.

$$M(p) = m_0 + m_1 p + m_2 p^2 + m_3 p^3$$

by Subsituting values of $m_0 \; m_1 \; m_2 \; m_3$.

$$M(p) = m_0 + m_2 p^2 \Rightarrow 1 + p^2$$

(ii) The Generator Polynomial is given by.

$$G(p) = 1 + p + p^3.$$

(iii) The non-systematic Cyclic code is given by.

$$X(p) = M(p) \cdot G(p)$$

$$= (1+p^2)(1+p+p^3)$$

$$= 1 + p + p^3 + p^2 + p^3 + p^5$$

$$= 1 + p + p^2 + (1 \oplus 1) p^3 + p^5.$$

$$\Rightarrow 1 + p + p^2 + 0p^3 + 0p^4 + p^5 + 0p^6$$

The degree of the code word polynomial is 6

i.e, $(n-1)$.

$$\boxed{X = (\underset{x_6 x_5 x_4}{1\,1\,1}\ \underset{x_3}{0}\ \underset{x_2}{0}\ \underset{x_1}{1}\ \underset{x_0}{0})}$$

$0\,1\,0\,0\,1\,1\,1$
$x^6 x^5 x^4 x^3 x^2 x^1 x^0$

This is the code word for the given msg word.

lly we can obtain the other non synthetic ~~code~~ code. words

Systematic Code ~~word~~ Vectors.

$n = 7, \; k = 4$

(i) $M(p) \times p^{n-k}$.

$$p^{n-k} M(p) = p^3(1+p^2).$$

$$= p^3 + p^5$$

(ii) divide $p^{n-k} M(p)$ by the generator polynomial.

$$\frac{p^{n-k} M(p)}{G(p)} = \frac{p^3+p^5}{1+p+p^3}$$

$$p^3+0p^2+p+1 \;\overline{)\; p^5+0p^4+p^3+0p^2+0p+0\;}\; (p^2+0p+0$$

$\uparrow$ $q(p)$ Quotient

$$1.p^5 + 0p^4 + p^3 + 0p^2 + 0p + 0$$

$$\oplus \quad \oplus \quad \oplus \quad \oplus$$

$$1.p^5 + 0p^4 + 0p^3 + 1.p^2 ..$$

$$0 \quad 0 \quad 0 \quad p^2 + 0p + 0.$$

$$\underbrace{p^2+0p+0}_{} \rightarrow \text{remainder.}$$

b(4)

$Q(p) = 0 + 0p + p^2.$

$C(p) = 0 + 0p + p^2.$ ← Represents the parity bits.

(ii) Obtain the codeword polynomial $x(p)$.

$$X(p) = [p^{n-k} M(p)] \oplus C(p).$$

$$= [0 + 0p + 0p^2 + p^3 + 0p^3 + p^5 + 0p^6] \oplus [0 + 0p + p^2]$$

$$= 0 + 0p + p^2 + p^3 + 0p^4 + p^5 + 0p^6$$

Code vector $= \underset{c_0\ \ c_1\ \ c_2\ \ m_0 m_1 m_2 m_3.}{[0 \cdot 0\ 1 : 1\,0\,1\,0]}$

= 1010 : 001

—— x ——.

Likewise obtain for all msg bits $m_3 m_2 m_1 m_0$.

SNo	Msg bits $m_3\ m_2\ m_1\ m_0$		Systematic codevectors $X = m_3\ m_2\ m_1\ m_0\ c_2\ c_1\ c_0$
1	0 0 0 0	→	0 0 0 0 0 0 0
2	0 0 0 1	→	0 0 0 1 0 1 1
3	0 0 1 0	→	0 0 1 0 1 1 0
4	0 0 1 1	→	0 1 0 0 0 0 1
5	0 1 0 0	→	0 1 0 0 1 1 1
6	0 1 0 1	→	0 1 0 1 1 0 0
7	0 1 1 0	→	0 1 1 0 0 0 1
8	0 1 1 1	→	0 1 1 1 0 1 0
9	1 0 0 0	→	1 0 0 0 1 0 1
10	1 0 0 1	→	1 0 0 1 1 1 0
11	1 0 1 0	→	1 0 1 0 0 1 1
12	1 0 1 1	→	1 0 1 1 0 0 0
13	1 1 0 0	→	1 1 0 0 0 1 0
14	1 1 0 1	→	1 1 0 1 0 0 1
15	1 1 1 0	→	1 1 1 0 1 0 0
16	1 1 1 1	→	1 1 1 1 1 1 1.

b(5)

Generator And parity check Matrices of cyclic codes.

(A) Non-Systematic form of Generator Matrix

Since some of cyclic codes are subclass of linear block codes, generator & parity check matrices can also be defined for cyclic codes

→ Generator Matrix has the size $K \times n$,

$K \rightarrow$ rows & $n \rightarrow$ columns. $\qquad q = n-K$

$$G(p) = p^{q} + g_{q-1}p^{q-1} + \dots + g_1 p + 1$$

multiply both sides ⇒ p^{i}

$$p^{i}G(p) = p^{i}(p^{q} + g_{q-1}p^{q-1} + \dots + g_1 p + 1)$$

$$= p^{i+q} + g_{q-1}p^{i+q-1} + \dots + g_1 p^{i+1} + p^{i}$$

$$i = (k-1), (k-2) \dots 2, 1, 0.$$

Example:

(1) Obtain the generator matrix corresponding to $G(p) = p^3 + p^2 + 1$ for a $(7, 4)$ cyclic code.

Sol^n. Here $n = 7$, $k = 4$, $q = 7 - 4 = 3$.

$$p^{i}G(p) \Rightarrow p^{i}(p^3 + p^2 + 1) \Rightarrow p^{i+3} + p^{i+2} + p^{i}$$

Since $k-1 = 3$, $i = 3, 2, 1, 0$.

Generator Matrix:

For row1 : $i = 3 \Rightarrow p^3 G(p) = p^6 + p^5 + p^3$

For row2 : $i = 2 \Rightarrow p^2 G(p) = p^5 + p^4 + p^2$

For row3 : $i = 1 \Rightarrow p\, G(p) = p^4 + p^3 + p$

For row4 : $i = 0 \Rightarrow G(p) = p^3 + p^2 + 1.$

Generator Matrix (n, k) code of size $k \times n$.

(4×7)

$$G_{4\times7} = \begin{array}{c} \\ R_1 \\ R_2 \\ R_3 \\ R_4 \end{array} \begin{array}{c} \begin{array}{ccccccc} P_6 & P_5 & P_4 & P_3 & P_2 & P_1 & P_0 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}_{4\times7} \end{array}$$

5(a) (b)

2Q $G(p) = p^3 + p + 1$; find code vectors

$$M = \underset{m_3 m_2 m_1 m_0}{(1001)} \qquad X = MG$$

Sol^n (i) To obtain generator Matrix.

$$p^i G(p) = p^{i+3} + p^{i+1} + p^i \qquad i = 3, 2, 1, 0. \text{ Since } (k-1) = 3$$

Row 1 : $i = 3 \Rightarrow p^3 G(p) = p^6 + p^4 + p^3$

Row 2 : $i = 2 \Rightarrow p^2 G(p) = p^5 + p^3 + p^2.$

Row 3 = $i = 1 \Rightarrow P\, G(p) = p^4 + p^2 + p^1$

Row 4 = $i = 0 \Rightarrow 1 \cdot G(p) = p^3 + p + 1.$

→ The above set of polynomials is transformed into a generator matrix of size 4×7 as shown below

$$G = \begin{array}{c} \\ \text{Row 1} \\ \text{Row 2} \\ \text{Row 3} \\ \text{Row 4} \end{array} \begin{array}{c} \begin{array}{ccccccc} p^6 & p^5 & p^4 & p^3 & p^2 & p^1 & p^0 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}_{4\times 7} \end{array}$$

Since cyclic code is a subclass of linear block code, its code vectors can be obtained as

$X = MG$

(ii) To obtain the code vectors:

Here M is the $1\times K$ msg vector & G is generator matrix. here $k=4$

msg vector is 4 bits

$M = m_3 \quad m_2 \quad m_1 \quad m_0 = 1001$

$$X = [1\ 0\ 0\ 1]\begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$= 1\oplus 0\oplus 0+0,\ 0\oplus 0\oplus 0\oplus 0,\ 1\oplus 0\oplus 0\oplus 0,$
$1\oplus 0\oplus 0\oplus 1,\ 0\oplus 0\oplus 0\oplus 0,\ 0\oplus 0\oplus 0\oplus 0 \oplus 1,$
$0\oplus 0\oplus 0\oplus 1.$

$= (1, 0, 1, 0, 0, 1, 1)$

(B). Systematic form of Generator Matrix

Systematic form of generator matrix is.

$$G = \left[I_k \mid P_{k \times q} \right]_{k \times n}$$

t^{th} row of this matrix will be represented by the polynomial form as

$$i^{th} \text{ row of } G = p^{n-t} + R_t(p) \;;\; t = 1, 2, 3 \ldots k$$

→ Lets divide p^{n-t} by a generator matrix $G(p)$.

$$\frac{p^{n-t}}{G(p)} = Quo + \frac{Rem}{G(p)}.$$

Let Remainder $= R_t(p)$,

Quotient $= Q_t(p)$.

$$\Rightarrow \frac{p^{n-t}}{G(p)} = Q_t(p) + \frac{R_t(p)}{G(p)}$$

$$p^{n-t} = Q_t(p)\, G(p) \oplus R_t(p) \quad \ldots t = 1, 2, \ldots k$$

It can be rewritten as

$$p^{n-t} \oplus R_t(p) = Q_t(p)\, G(p).$$

16 Parity check Matrix:

Once the generator matrix in systematic form is obtained then parity check matrix can be obtained. ~~per~~

3Q Find out the generator matrix for a systematic (7,4) cyclic code if $G(p) = p^3+p+1$. Find out the parity check matrix.

Sol^n (I) To obtain generator polynomial

$$t^{th} \text{ row} \Rightarrow p^{n-t} + R_t(p) = Q_t(p)\,G(p).$$

$t = 1, 2, \ldots k$

$n=7,\quad k=4 \qquad q = n-k = 3.$

$$p^{7-t} + R_t(p) = Q_t(p)\left[p^3+p+1\right] \quad \forall\ t = 1,2,3,4$$

$$t=1 \Rightarrow p^6 + R_t(p) = Q_t(p)\,(p^3+p+1).$$

(a) To obtain $R_t(p)$ and $Q_t(p)$ for 1st row.

RHS, LHS represents 1st row of systematic Generator matrix.

$Q_t(p) = ?$

p^{n-t} by $G(p) \Rightarrow p^6$ by $G(p) = p^3+p+1$.

$$\begin{array}{l}
p^3+p+1 \,\big)\, p^6+0+0 \,\big(\, p^3+p+1 \leftarrow \text{Quotient} \\
\qquad\qquad\ \ p^6 + p^4 + p^3 \\
\qquad\qquad\ \ \oplus \quad \oplus \quad \oplus \\
\hline
\qquad\qquad\ \ 0 + p^4 + p^3 + 0 + 0 \\
\qquad\qquad\quad\ \ p^4 + 0 + p^2 + p \\
\hline
\qquad\qquad\qquad\quad p^3 + p^2 + p + 0 \\
\qquad\qquad\qquad\quad p^3 + 0 + p + 1 \\
\hline
\qquad\qquad\qquad\qquad\qquad p^2+1 \leftarrow \text{Remainder.}
\end{array}$$

$Q_t(p) = p^3 + p + 1$; $R_t(p) = p^2 + 1$.

At $t = 1 \Rightarrow Q_t(p)(p^3 + p + 1) = p^6 + R_t(p)$

$Q_t(p)(p^3+p+1) = (p^3+p+1)(p^3+p+1)$

RHS $\Rightarrow$

$$= p^6 + p^4 + p^3 + p^4 + p^2 + p + p^3 + p + 1$$

$$= p^6 + p^4(1 \oplus 1) + p^3(1 \oplus 1) + p^2 + p(1 \oplus 1) + 1$$

$$= p^6 + p^4(0) + p^3(0) + p^2 + p(0) + 1.$$

$$= p^6 + p^2 + 1.$$

LHS $\Rightarrow p^6 + (p^3 + 1)$

$\Rightarrow \boxed{LHS = RHS}$ Row 1 $= p^6 + p^2 + 1$.

(ii). Other Rows polynomials:

By using the same procedures,

$t = 2 = 2^{nd}$ row $= p^5 + p^2 + p + 1$

$t = 3 = 3^{rd}$ row $= p^4 + p^2 + p$

$t = 4 \Rightarrow 4^{th}$ row $= p^3 + p + 1$.

(iii) Conversion of row polynomial into matrix:

Above eqns can be transformed to

$$G = \begin{array}{c} R1 \\ R2 \\ R3 \\ R4 \end{array} \left[\begin{array}{cccc:ccc} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{array}\right]_{4 \times 7}$$

This is the required generator matrix in systematic form. Codevector $X = MG$.

$$M = \overset{m_3 m_2 m_1 m_0}{1100}$$

$$X = \begin{bmatrix} 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$X = (1\ 1\ 0\ 0\ 0\ 1\ 0)$$

(ii) Parity Check Matrix:

$$G = \left[I_k : P_{k \times q} \right]_{k \times n}$$

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}_{4 \times 3}$$

$$H = \left[P^T : I_q \right]_{q \times n}$$

$q = n - k$ / $H \rightarrow P^T I_q$

P^T is submatrix. I_q = identity.

By taking transpose of 'P',

$$H = \begin{bmatrix} \underbrace{1\ 1\ 1\ 0}_{P^T} & \underbrace{1\ 0\ 0}_{I_3} \\ 0\ 1\ 1\ 1 & 0\ 1\ 0 \\ 1\ 1\ 0\ 1 & 0\ 0\ 1 \end{bmatrix}_{3 \times 7}$$

— x —

CRC

Cyclic Redunancy Check

Error detecting code. usedin digital Networks

CRC is Simple to implement in

Easy to analyze.

Good at detecting errors.

Example:

The message 11001001 is to be transmitted using the CRC polynomial x^3+1 to protect it from errors. The msg. that should be transmitted is ___

Sol Given msg: 11001001, CRC polynomial $= x^3+1$

x^3 x^2 x x^0

CRC Generator = 1 0 0 1

Since its order is '3' add (append) 3 zero's at the end

Now the new msg = 11001001000

```
11001001000
⊕1001
 01011
  1001
 001000
   1001
 0001100
    1001
 00001010
     1001
      011  reminder
```

The msg transmitted is

11001001011

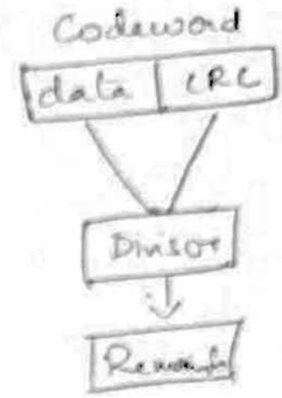
Q Explain the Generate the CRC Code for the data word of 11001 0101 & the divisor is 10101

Sol

msg: 110010101

divisior = 10101

$x_4 \ x_3 \ x_2 \ x_1 \ x_0$
$1 \ 0 \ 1 \ 0 \ 1$



Generator polynomial: $x^4 + x^2 + 1$

Highest order is 4 so append 4 zero's to the given msg to find the complete codeword.

```
110010101 0000   ( 111110111
10101
-----
 11000
 10101
 -----
 011011
  10101
  -----
  011100
   10101
   -----
   010011
    10101
    -----
    0011000
      10101
      -----
      11010
      10101
      -----
      11110
      10101
      -----
       1011 ← CRC.
```

Transmitted msg is 11001 01011011

— x —